

COMMUTATIVE ALGEBRA EXAM

1. IRREDUCIBLE POLYNOMIALS AND GALOIS THEORY

Exercise 1.1. Let $k = \mathbb{C}(t)$ be the field of rational functions on $\mathbb{C}$. Prove that for any n there exists an irreducible polynomial $P(t) \in k[t]$ of degree n .

Proof. Let $P(x) \in k[x] = \mathbb{C}(t)[x]$ be $P(x) = x^n - t$. This polynomial has no root in $k = \mathbb{C}(t)$, for if $(f/g)^n = t$ then $n(\deg f - \deg g) = 1$, which is impossible for $n > 1$. But unfortunately this doesn't prove that it's irreducible: in $\mathbb{R}[x]$, the polynomial $(x^2 + 1)(x^2 + 2) = x^4 + 3x^2 + 2$ is reducible and has no roots.

To prove $x^n + t$ is irreducible we shall use the Gauss lemma. Forget the above notation and suppose that $fg = x^n + t$, where $f, g \in \mathbb{C}(t)[x]$. We want to clear the denominators in the coefficients of f and g , so we multiply by the product D of all such denominators and obtain

$$f_1 g_1 = (x^n + t)D,$$

where $D \in \mathbb{C}[t]$ and $f_1, g_1 \in \mathbb{C}[t][x]$. Now we apply Gauss lemma, that is, the content of the product is the product of the contents. Since $x^n + t$ is primitive (i.e. has content 1), then the content of $f_1 g_1$ equals the content of D , but D is a constant (constants here are polynomials in $\mathbb{C}[t]$). We obtain

$$FG = x^n + t$$

where F and G are the primitive parts of f_1 and g_1 . This holds up to multiplication by a complex number.

Now we reduce modulo t and obtain

$$[F][G] = [x^n],$$

and conclude that, as classes,

$$[F] = [cx^m], \quad [G] = [c^{-1}x^{n-m}].$$

Thus we see that each of F, G lost their constant term when we reduced, i.e. that their constant terms are divisible by t . This means that t^2 divides the constant term of the product FG . But this is impossible, since the constant term of FG is t .

The only way this makes sense is if $m = 0$ or $m = n$. To conclude we note that reduction modulo t preserves the degree of both F , and G because their leading coefficients multiply to 1. Thus $m = 0$ or $m = n$ imply that F or G is a unit, which in turn implies so for f or g . $\square$

Exercise 1.2. Prove that for any $d \in \mathbb{Z}^{>0}$ there exists an irreducible polynomial $P \in \mathbb{C}[x, y]$ such that $P(x, y) = 0$ defines a smooth submanifold in $\mathbb{C}^2$.

Proof. $x^n + y$ since $\partial_y = 1$. This is an irreducible polynomial because y is irreducible. More precisely, if $x^n + y = fg$, then either of f, g must have no y variable, say $f \in \mathbb{C}[x]$, and the other one must be degree 1 in y . Suppose $g = yg' + g''$ for $g', g'' \in \mathbb{C}[x]$. But then $x^n + y = fg = yfg' + fg''$, which means that f cannot have positive degree in x , but $f \in \mathbb{C}[x]$, so it is a unit. $\square$

2. TENSOR PRODUCT OF R-MODULES

Exercise 2.1. Let A be a ring without zero divisors, and M an A -module which is flat over all local rings $A_{\mathfrak{p}} \supset A$ obtained by localizing A at prime ideals. Prove that M is flat.

Proof. Let

$$0 \longrightarrow M_1 \longrightarrow M_2 \longrightarrow M_3 \longrightarrow 0$$

be exact. Then consider

$$0 \longrightarrow M_1 \otimes_A M \xrightarrow{\varphi} M_2 \otimes_A M \longrightarrow M_3 \otimes_A M \longrightarrow 0.$$

We need to show that the map $\varphi : M_1 \otimes_A M \rightarrow M_2 \otimes_A M$ is injective. So basically the claim is: if that map is injective over all localizations $A_{\mathfrak{p}}$ then it is injective over A .

So it's basically two steps: (1) show that

$$(2.1.1) \quad (\text{Ker } \varphi)_{\mathfrak{p}} = \text{Ker } \varphi^{\mathfrak{p}}$$

where $\varphi^{\mathfrak{p}}$ is the induced map over $A_{\mathfrak{p}}$ for any $\mathfrak{p} \subset A$ prime. More precisely:

$$0 \longrightarrow (M_1)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \xrightarrow{\varphi^{\mathfrak{p}}} (M_2)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \longrightarrow (M_3)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}} \longrightarrow 0$$

where the localizations of each module are given by tensoring with $A_{\mathfrak{p}}$ over A , e.g. $M_{\mathfrak{p}} = A_{\mathfrak{p}} \otimes_A M$ by definition. Thus the above sequence is exactly

$$0 \longrightarrow (M_1 \otimes_A M)_{\mathfrak{p}} \xrightarrow{\varphi^{\mathfrak{p}}} (M_2 \otimes_A M)_{\mathfrak{p}} \longrightarrow (M_3 \otimes_A M)_{\mathfrak{p}} \longrightarrow 0$$

and we see that Equation 2.1.1 follows by definition. Indeed, both sets are barely

$$\left\{ \frac{a}{b} x : \varphi(x) = 0, \frac{a}{b} \in A_{\mathfrak{p}} \right\}.$$

By the exercise's hypothesis, we know that $\text{Ker } \varphi^{\mathfrak{p}} = 0$.

Then, step (2) is: show that if any A -module K is zero as an $A_{\mathfrak{p}}$ -module for any $\mathfrak{p} \subset A$, then it is zero as an A -module. Here's AI's proof: consider the annihilator of an element $m \in M$, it's the set of scalars that kill it, i.e. $\text{Ann}(m) = \{a \in A : am = 0\}$. It's nonempty by hypothesis (yes, the fact that m is zero over $A_{\mathfrak{p}}$ literally means that there is $a \notin \mathfrak{p}$ such that $am = 0$). This just says that $\text{Ann}(m) \not\subset \mathfrak{p}$ for any $\mathfrak{p}$. But that's not true: any ideal is contained in some maximal (and thus prime) ideal. The reason for that is Zorn's lemma: the set of ideals containing $\text{Ann}(m)$ is inductive (i.e. every chain is bounded, in this case by the union of the ideals), so it has a maximal element, and maximal ideals are prime because the quotient is a field, which is in particular a domain. □

Definition 2.2. A *boolean ring* is a ring such that all its elements are idempotents, that is, satisfy $a^2 = a$.

Exercise 2.3. Prove that any finitely generated module over a Boolean ring is flat.

Proof. The claim is false. Consider the boolean ring $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, equipped with the usual ring structure for direct sums which is given by applying the operations entry by entry.

This module has torsion:

$$([1], [0])([0], [1]) = ([0], [0]).$$

Ah! But this ring is not a domain. Forget it.

□